Documentar cómo verificaron que systemd se encuentra activo

Systemd como primer proceso

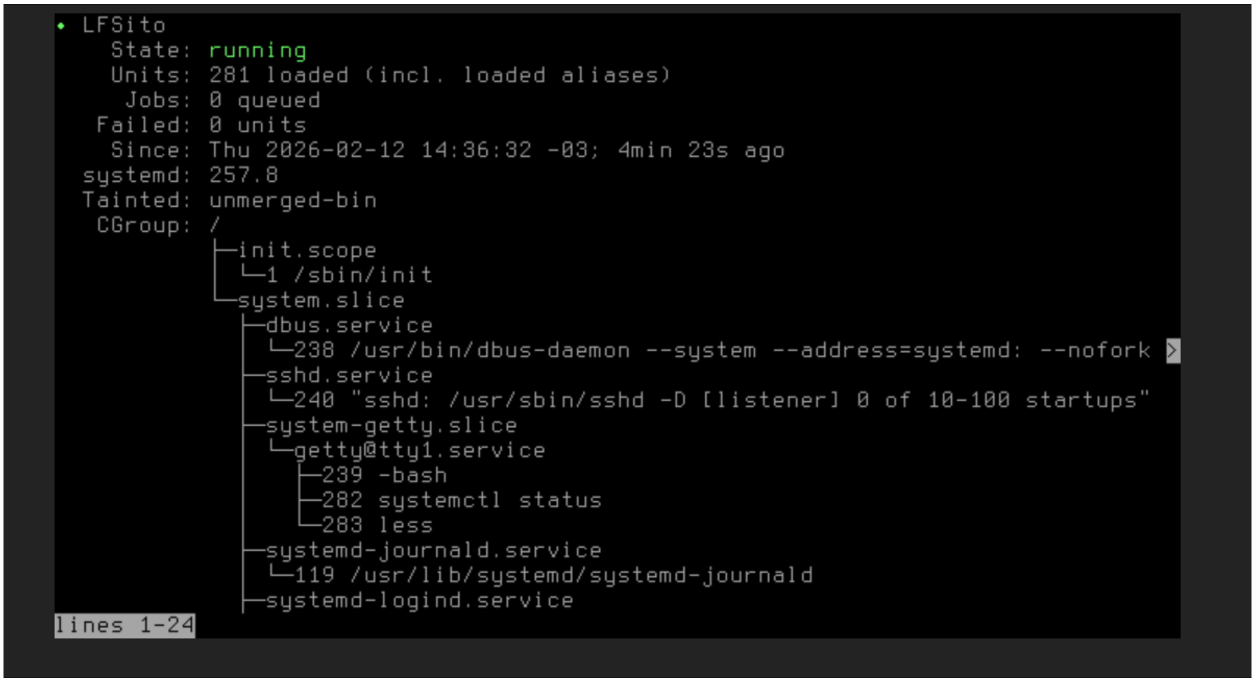
ps -p 1 -o pid,comm,args

```
-bash-5.3# ps -p 1 -o pid,comm,args
    PID COMMAND         COMMAND
      1 systemd         /sbin/init
-bash-5.3#
```

pstree

```
-bash-5.3# pstree
systemd─┬─bash───pstree
        ├─dbus-daemon
        ├─sshd
        ├─systemd-journal
        ├─systemd-logind
        ├─systemd-network
        ├─systemd-nsresou───5*[systemd-nsresou]
        ├─systemd-oomd
        ├─systemd-resolve
        ├─systemd-timesyn───{systemd-timesyn}
        └─systemd-udevd
-bash-5.3# _
```

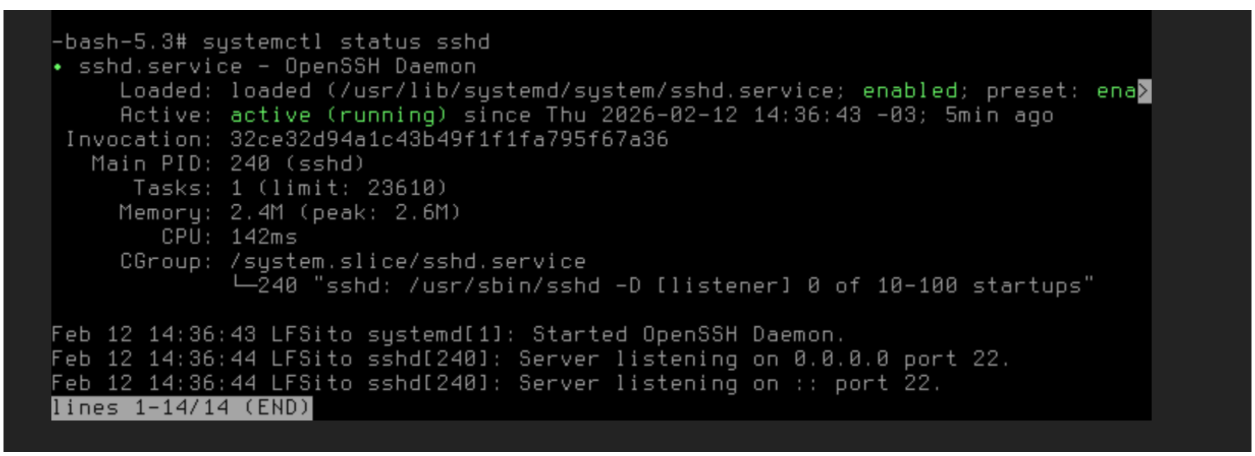
systemctl status



```
• LFSito
    State: running
    Units: 281 loaded (incl. loaded aliases)
     Jobs: 0 queued
   Failed: 0 units
    Since: Thu 2026-02-12 14:36:32 -03; 4min 23s ago
  systemd: 257.8
  Tainted: unmerged-bin
   CGroup: /
           ├─init.scope
           │ └─1 /sbin/init
           └─system.slice
             ├─dbus.service
             │ └─238 /usr/bin/dbus-daemon --system --address=systemd: --nofork >
             ├─sshd.service
             │ └─240 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
             ├─system-getty.slice
             │ └─getty@tty1.service
             │   ├─239 -bash
             │   ├─282 systemctl status
             │   └─283 less
             ├─systemd-journald.service
             │ └─119 /usr/lib/systemd/systemd-journald
             ├─systemd-logind.service
lines 1-24
```

Mostrar al menos un servicio levantado/gestionado con systemd

systemctl status sshd



```
-bash-5.3# systemctl status sshd
• sshd.service - OpenSSH Daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-02-12 14:36:43 -03; 5min ago
 Invocation: 32ce32d94a1c43b49f1f1fa795f67a36
   Main PID: 240 (sshd)
      Tasks: 1 (limit: 23610)
     Memory: 2.4M (peak: 2.6M)
        CPU: 142ms
     CGroup: /system.slice/sshd.service
             └─240 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 12 14:36:43 LFSito systemd[1]: Started OpenSSH Daemon.
Feb 12 14:36:44 LFSito sshd[240]: Server listening on 0.0.0.0 port 22.
Feb 12 14:36:44 LFSito sshd[240]: Server listening on :: port 22.
lines 1-14/14 (END)
```

1

Logs

journalctl -u sshd

```
Jan 24 15:31:28 LFSito systemd[1]: Started OpenSSH Daemon.
Jan 24 15:31:29 LFSito sshd[235]: Server listening on 0.0.0.0 port 22.
Jan 24 15:31:29 LFSito sshd[235]: Server listening on :: port 22.
-- Boot 0da106bf1133462c9e7c07e2308c3b2e --
Jan 24 15:39:15 LFSito systemd[1]: Started OpenSSH Daemon.
Jan 24 15:39:16 LFSito sshd[236]: Server listening on 0.0.0.0 port 22.
Jan 24 15:39:16 LFSito sshd[236]: Server listening on :: port 22.
Jan 24 15:45:56 LFSito sshd-session[275]: Received disconnect from 192.168.0.11>
Jan 24 15:45:56 LFSito sshd-session[275]: Disconnected from user asdy 192.168.0>
-- Boot a79ce60593f8492a82ca5b338345a823 --
Jan 31 10:08:50 LFSito systemd[1]: Started OpenSSH Daemon.
Jan 31 10:08:51 LFSito sshd[235]: Server listening on 0.0.0.0 port 22.
Jan 31 10:08:51 LFSito sshd[235]: Server listening on :: port 22.
-- Boot fadeafed690f442694faa3e12dc9e7e0 --
Feb 04 20:17:44 LFSito systemd[1]: Started OpenSSH Daemon.
Feb 04 20:17:44 LFSito sshd[230]: Server listening on 0.0.0.0 port 22.
Feb 04 20:17:44 LFSito sshd[230]: Server listening on :: port 22.
-- Boot 4c1061a7ce1d48afa23b7f20ab93bd88 --
Feb 06 12:06:30 LFSito systemd[1]: Started OpenSSH Daemon.
Feb 06 12:06:31 LFSito sshd[238]: Server listening on 0.0.0.0 port 22.
Feb 06 12:06:31 LFSito sshd[238]: Server listening on :: port 22.
-- Boot 1559bf12789c49498dee871603ce92bb --
Feb 06 12:15:52 LFSito systemd[1]: Started OpenSSH Daemon.
Feb 06 12:15:52 LFSito sshd[238]: Server listening on 0.0.0.0 port 22.
lines 1-24
```

Configuración de la unidad

cat /lib/systemd/system/sshd.service

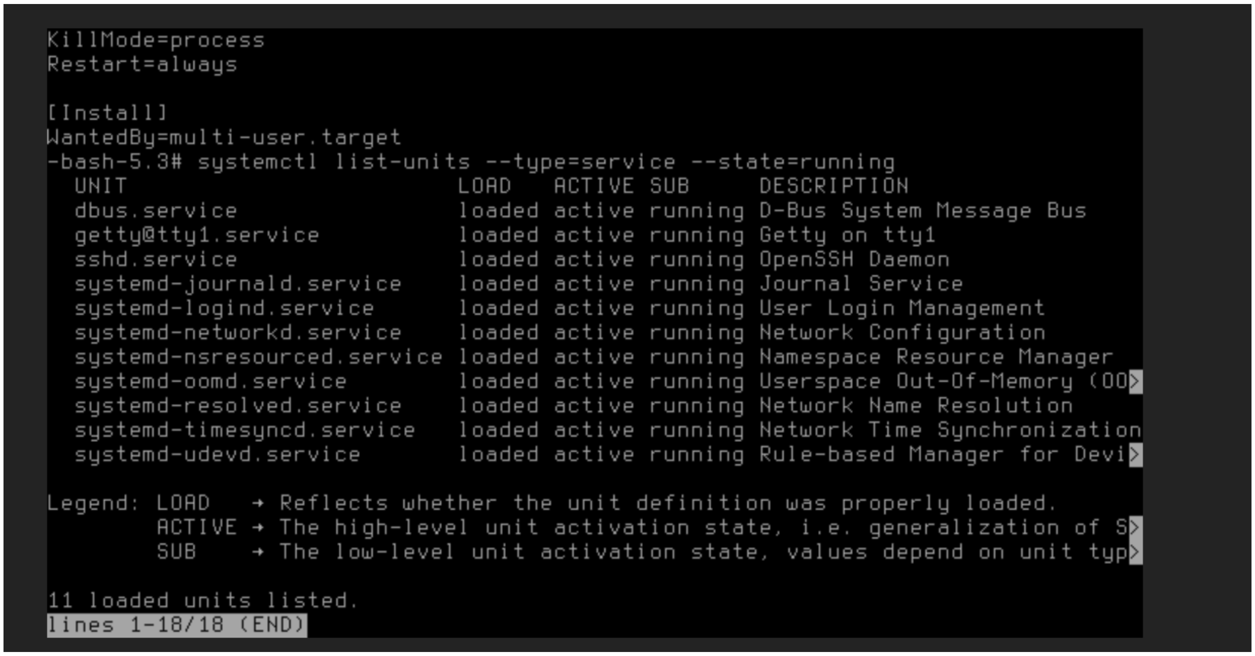
```
-bash-5.3# cat /lib/systemd/system/sshd.service
[Unit]
Description=OpenSSH Daemon

[Service]
ExecStart=/usr/sbin/sshd -D
ExecReload=/bin/kill -HUP $MAINPID
KillMode=process
Restart=always

[Install]
WantedBy=multi-user.target
-bash-5.3#
```

Lista de servicios activos

systemctl list-units --type=service --state=running



```
KillMode=process
Restart=always

[Install]
WantedBy=multi-user.target
-bash-5.3# systemctl list-units --type=service --state=running
  UNIT                         LOAD   ACTIVE SUB     DESCRIPTION
  dbus.service                 loaded active running D-Bus System Message Bus
  getty@tty1.service           loaded active running Getty on tty1
  sshd.service                 loaded active running OpenSSH Daemon
  systemd-journald.service     loaded active running Journal Service
  systemd-logind.service       loaded active running User Login Management
  systemd-networkd.service     loaded active running Network Configuration
  systemd-nsresourced.service  loaded active running Namespace Resource Manager
  systemd-oomd.service         loaded active running Userspace Out-Of-Memory (OO>
  systemd-resolved.service     loaded active running Network Name Resolution
  systemd-timesyncd.service    loaded active running Network Time Synchronization
  systemd-udevd.service        loaded active running Rule-based Manager for Devi>

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of S>
        SUB    → The low-level unit activation state, values depend on unit typ>

11 loaded units listed.
lines 1-18/18 (END)
```

Sytemd vs Sysvinit

Al utilizar systemd como sistema init en este proyecto de LFS, se puede contrastar su funcionamiento con lo que ocurriría si se hubiera utilizado Sysvinit tradicional. La diferencia más evidente se observa en el proceso de arranque: con systemd, los servicios se inician de manera paralela, lo que en teoría reduce significativamente el tiempo de boot, mientras que con Sysvinit cada servicio esperaría a que el anterior terminara de iniciar antes de comenzar. Sobre la gestión de servicios, systemd centraliza todo mediante el comando systemctl, permitiendo iniciar, detener o verificar el estado de cualquier servicio con este. En cambio, Sysvinit requeriría ejecutar scripts individuales ubicados en /etc/init.d/ con diferentes implementaciones según cada servicio. El manejo de logs también difiere notablemente. Systemd implementa journalctl para consultar todos los registros del sistema desde este mismo. Las dependencias entre servicios en systemd se declaran explícitamente en los archivos .service mediante directivas como After= o Requires=, mientras que en Sysvinit estas dependencias se manejarían mediante números de secuencia (S01, S02, etc.) que resultan menos claros y más propensos a errores de configuración. Aunque Sysvinit es conceptualmente más simple por basarse en scripts de shell convencionales, systemd ofrece capacidades más robustas para sistemas modernos que requieren mayor control y eficiencia en la gestión de servicios.

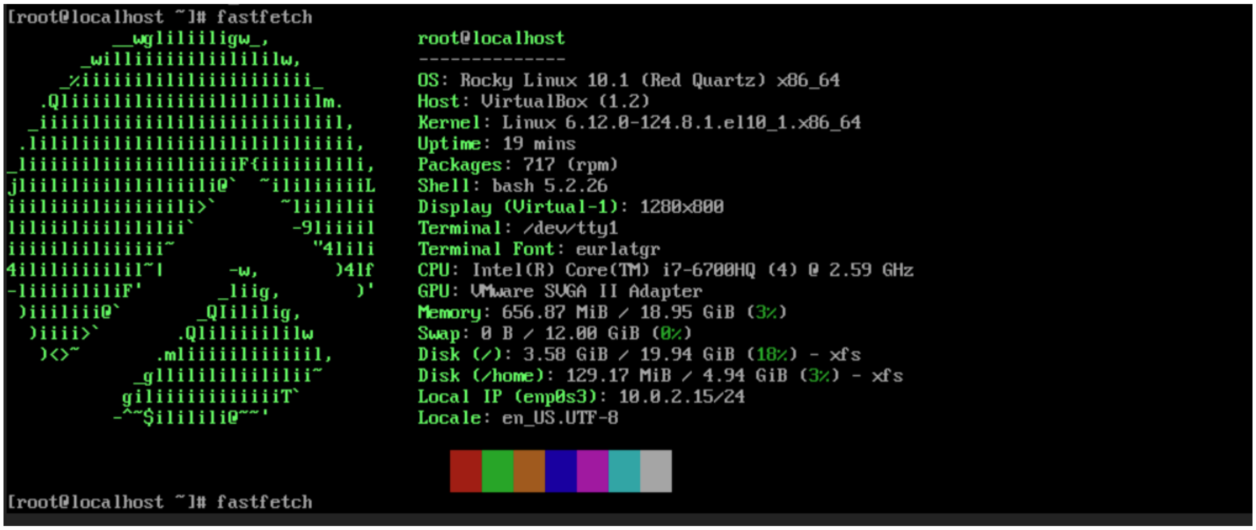
Host Capturas

Capturas Sobre información del Host

```
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   4
  On-line CPU(s) list:    0-3
Vendor ID:                GenuineIntel
  Model name:             Intel(R) Core(TM) i7-6700HQ CPU @ 2.60GHz
    BIOS Model name:        CPU @ 0.0GHz
    BIOS CPU family:      0
    CPU family:           6
    Model:                94
    Thread(s) per core:   1
    Core(s) per socket:   4
    Socket(s):            1
    Stepping:             3
    BogoMIPS:             5184.00
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_
                          tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 movbe popcnt aes xsave
                           avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti fsgsbase bmi1 avx2 bmi2 invpcid rdseed adx clflushopt arat md_clear flush
                          _l1d arch_capabilities
Virtualization features:
  Hypervisor vendor:      KVM
  Virtualization type:    full
Caches (sum of all):
  L1d:                    128 KiB (4 instances)
  L1i:                    128 KiB (4 instances)
  L2:                     1 MiB (4 instances)
  L3:                     24 MiB (4 instances)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-3
Vulnerabilities:
  Gather data sampling:   Unknown: Dependent on hypervisor status
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:          Not affected
  L1tf:                   Mitigation; PTE Inversion
  Mds:                    Mitigation; Clear CPU buffers; SMT Host state unknown
  Meltdown:               Mitigation; PTI
  Mmio stale data:        Mitigation; Clear CPU buffers; SMT Host state unknown
  Reg file data sampling: Not affected
  Retbleed:               Vulnerable
  Spec rstack overflow:   Not affected
  Spec store bypass:      Vulnerable
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Retpolines; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected; BHI Retpoline
  Srbds:                  Unknown: Dependent on hypervisor status
  Tsa:                    Not affected
  Tsx async abort:        Not affected
[root@localhost ~]#
```

```
[root@localhost ~]# free -h
               total        used        free      shared  buff/cache   available
Mem:            18Gi       555Mi        18Gi        10Mi       198Mi        18Gi
Swap:           11Gi          0B        11Gi
[root@localhost ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb5        20G  3.2G   17G  16% /
devtmpfs        9.5G     0  9.5G   0% /dev
tmpfs           9.5G     0  9.5G   0% /dev/shm
tmpfs           3.8G   11M  3.8G   1% /run
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
/dev/sdb2       960M  362M  599M  38% /boot
/dev/sdb4       5.0G  130M  4.9G   3% /home
tmpfs           1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           1.9G  4.0K  1.9G   1% /run/user/0
[root@localhost ~]#
```

fastfetch (en vez de neofetch)



Sdb6 siendo la partición del LFS